Model Derivations

Companion notes to "Trade Tariffs and Exchange Rates"

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These notes derive three models: the two-country benchmark, the two-layer nested three-country model with the reversal threshold, and the canonical competing-exporters model from the preferential-trade-agreement (PTA) literature. Prose is limited to defining variables and linking equations.

1 Two-country model

1.1 Setup

Countries $i \in \{A, B\}$. Goods: tradable T_i (produced only in i, consumed in both) and non-tradable N_i. Labor is the only factor, immobile across countries, endowment L_i.

Technology (linear):

$$Y_{T_i} = A_{T_i} L_{T_i}, \qquad Y_{N_i} = A_{N_i} L_{N_i}, \qquad L_{T_i} + L_{N_i} = L_i. \tag{1}$$

Perfect competition and the normalization $P_{T_i} = 1$ (producer price of each tradable in its own currency) give

$$w_i = A_{T_i}, \qquad P_{N_i} = \frac{w_i}{A_{N_i}} = \frac{A_{T_i}}{A_{N_i}}. \tag{2}$$

Let e be the nominal exchange rate: units of A's currency per unit of B's currency. The price of T_B in A's currency is e; the price of T_A in B's currency is $1/e$.

Preferences (Cobb–Douglas, common weights):

$$U_i = C_{T_A i}^{\alpha_{T_A}}\, C_{T_B i}^{\alpha_{T_B}}\, C_{N_i}^{\alpha_N}, \qquad \alpha_{T_A} + \alpha_{T_B} + \alpha_N = 1, \tag{3}$$

where C_{gi} is country i's consumption of good g.

1.2 Demands

Cobb–Douglas expenditure shares are constant. With income I_i and consumer prices $P_g^{(i)}$ (in i's currency),

$$P_g^{(i)} C_{gi} = \alpha_g I_i \quad \Longrightarrow \quad C_{gi} = \frac{\alpha_g I_i}{P_g^{(i)}}. \tag{4}$$

Under free trade $I_i = w_i L_i$, $P_{T_B}^{(A)} = e$, $P_{T_A}^{(B)} = 1/e$.

1.3 Equilibrium under free trade

Country A's trade balance in A's currency is export revenue minus import spending. From (4), B's purchases of T_A are $C_{T_A B} = \alpha_{T_A} I_B/(1/e) = \alpha_{T_A} w_B L_B\, e$, valued at producer price 1 in A's currency, and A's purchases of T_B are $C_{T_B A} = \alpha_{T_B} w_A L_A/e$, valued at price e. Therefore

$$\mathrm{TB}_A = 1 \cdot C_{T_A B} - e\, C_{T_B A} = \alpha_{T_A} w_B L_B\, e - \alpha_{T_B} w_A L_A. \tag{5}$$

Setting $\mathrm{TB}_A = 0$ (by Walras' law this is the only independent condition):

$$\boxed{e = \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{w_A L_A}{w_B L_B} = \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{A_{T_A} L_A}{A_{T_B} L_B}} \tag{6}$$

1.4 Tariff

Country A levies ad valorem tariff $\tau \geq 0$ on imports of T_B. The consumer price in A becomes $(1+\tau)e$; producers in B still receive 1. Revenue is rebated lump-sum:

$$I_A = w_A L_A + T_A^{\mathrm{rev}}, \qquad T_A^{\mathrm{rev}} = \tau e\, C_{T_B A}. \tag{7}$$

Using $C_{T_B A} = \alpha_{T_B} I_A/[(1+\tau)e]$,

$$I_A = w_A L_A + \frac{\tau}{1+\tau}\, \alpha_{T_B} I_A \quad \Longrightarrow \quad I_A = \frac{w_A L_A}{1 - \dfrac{\alpha_{T_B}\tau}{1+\tau}}. \tag{8}$$

The trade balance (flows valued at producer prices; the tariff wedge is a domestic transfer):

$$\mathrm{TB}_A = C_{T_A B} - e\, C_{T_B A} = \alpha_{T_A} w_B L_B\, e \; - \; \frac{\alpha_{T_B} I_A}{1+\tau}. \tag{9}$$

Setting $\mathrm{TB}_A = 0$ and substituting (8):

$$\boxed{e(\tau) = \frac{1}{1+\tau} \cdot \frac{1}{1 - \dfrac{\alpha_{T_B}\tau}{1+\tau}} \cdot \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{A_{T_A} L_A}{A_{T_B} L_B}} \tag{10}$$

Result 1.1. *$e(\tau)$ is strictly decreasing in τ for $\alpha_{T_B} \in (0,1)$: a tariff appreciates the tariff-imposing country's currency.*

Proof. Write $e(\tau) = e(0) \cdot f(\tau)$ with $f(\tau) = \big[(1+\tau) - \alpha_{T_B}\tau\big]^{-1} = \big[1 + (1-\alpha_{T_B})\tau\big]^{-1}$. Since $1 - \alpha_{T_B} > 0$, $f' < 0$. $\square$

1.5 Relative wage, terms of trade, and real exchange rate

Relative wage. Wages in own currency are $w_A = A_{T_A}$ and $w_B = A_{T_B}$, both constant. The relative wage in a common currency (A's) is

$$\omega \;\equiv\; \frac{e\,w_B}{w_A} \;=\; \frac{A_{T_B}}{A_{T_A}}\,e \;\propto\; e. \tag{11}$$

A rise in e (depreciation of A's currency) is a fall in A's wage relative to B's.

Terms of trade. A's terms of trade are the price of its exports relative to its imports, in a common currency and at world (pre-tariff) prices:

$$\mathrm{ToT}_A \;\equiv\; \frac{P_{T_A}}{e\,P_{T_B}} \;=\; \frac{1}{e}. \tag{12}$$

Combining (11) and (12): the exchange rate, the (inverse) common-currency relative wage, and the (inverse) terms of trade are the same object up to constants,

$$e \;\propto\; \frac{w_B\,e}{w_A} \;\propto\; \frac{1}{\mathrm{ToT}_A}. \tag{13}$$

This holds because producer prices are unit labor costs ($P_{T_i} = w_i/A_{T_i} = 1$): one factor and constant returns make the identification exact.

Real exchange rate. The consumer price index in each country is the Cobb–Douglas index over consumer prices (constants κ_i collect the share terms):

$$P_A = \kappa_A \left[P_{T_A}\right]^{\alpha_{T_A}} \left[(1+\tau)e\right]^{\alpha_{T_B}} \left[P_{N_A}\right]^{\alpha_N} = \kappa_A \left[(1+\tau)e\right]^{\alpha_{T_B}} \left(\tfrac{A_{T_A}}{A_{N_A}}\right)^{\alpha_N}, \tag{14}$$

$$P_B = \kappa_B \left[1/e\right]^{\alpha_{T_A}} \left[P_{T_B}\right]^{\alpha_{T_B}} \left[P_{N_B}\right]^{\alpha_N} = \kappa_B\, e^{-\alpha_{T_A}} \left(\tfrac{A_{T_B}}{A_{N_B}}\right)^{\alpha_N}. \tag{15}$$

The real exchange rate is

$$q \;\equiv\; \frac{e\,P_B}{P_A} \;=\; \mathrm{const}\times e^{\,1-\alpha_{T_A}-\alpha_{T_B}}\,(1+\tau)^{-\alpha_{T_B}} \;=\; \mathrm{const}\times e^{\,\alpha_N}\,(1+\tau)^{-\alpha_{T_B}}, \tag{16}$$

so in logs

$$d\log q \;=\; \alpha_N\, d\log e \;-\; \alpha_{T_B}\,\frac{d\tau}{1+\tau}. \tag{17}$$

Equation (17) has three implications. First, q moves with e, attenuated by the non-tradable share α_N. Second, the tariff appreciates A in real terms directly, at any e, because it raises A's consumer price index. Third, since $P_{N_i} = A_{T_i}/A_{N_i}$ is a productivity ratio, non-tradables affect the level of q but enter its response to trade shocks only through the exponent α_N.

1.6 The Marshall–Lerner condition

Let export volume depend on the price foreign buyers face, $p^* = 1/e$, and import volume on the price home buyers face, $p = (1+\tau)e$. Write $X(p^*)$ and $M(p)$ with elasticities

$$\varepsilon_X \equiv -\frac{X'(p^*)\,p^*}{X} \;\geq\; 0, \qquad \varepsilon_M \equiv -\frac{M'(p)\,p}{M} \;\geq\; 0. \tag{18}$$

The trade balance in A's currency, at producer prices, is

$$\mathrm{TB}_A(e,\tau) = X(1/e) \;-\; e\,M\big((1+\tau)e\big). \tag{19}$$

Response to the exchange rate. Differentiate (19) with respect to e at $\tau = 0$:

$$\frac{\partial \mathrm{TB}_A}{\partial e} = -\frac{X'(1/e)}{e^2} - M - e\,M'(e) = \frac{\varepsilon_X X}{e} - M + \varepsilon_M M, \tag{20}$$

using $X' = -\varepsilon_X X e$ and $eM' = -\varepsilon_M M$. Evaluated at balanced trade ($X = eM$):

$$\boxed{\frac{\partial \mathrm{TB}_A}{\partial e} = M\big(\varepsilon_X + \varepsilon_M - 1\big)} \tag{21}$$

A depreciation improves the trade balance if and only if $\varepsilon_X + \varepsilon_M > 1$. This is the Marshall–Lerner condition. The term -1 in (21) is the valuation effect: a depreciation raises the home-currency cost of each unit of imports, which worsens the balance one-for-one. The two volume responses must offset this effect.

Response to the tariff. Differentiate (19) with respect to τ at fixed e:

$$\frac{\partial \mathrm{TB}_A}{\partial \tau} = -e\,M'\big((1+\tau)e\big)\,e = \frac{\varepsilon_M}{1+\tau}\,eM \;>\; 0 \quad \text{whenever } \varepsilon_M > 0. \tag{22}$$

A tariff improves the trade balance at a fixed exchange rate as long as import demand slopes down. (With a revenue rebate, income effects modify (22); in the Cobb–Douglas model of Section 1.4 the producer-price import value is $\alpha_{T_B} I_A/(1+\tau) \propto [1 + (1-\alpha_{T_B})\tau]^{-1}$, still decreasing in τ.)

Why Marshall–Lerner is necessary and sufficient for the conventional wisdom. Equilibrium requires $\mathrm{TB}_A(e(\tau),\tau) = 0$. By the implicit function theorem,

$$\frac{de}{d\tau} = -\frac{\partial \mathrm{TB}_A/\partial \tau}{\partial \mathrm{TB}_A/\partial e}. \tag{23}$$

The numerator is positive by (22): the tariff creates a surplus at the initial exchange rate. To restore balance, the exchange rate must move in the direction that worsens the trade balance. If Marshall–Lerner holds, that direction is appreciation, $de < 0$, which is the conventional result. If Marshall–Lerner fails, depreciation worsens the balance instead, and the same tariff-induced surplus requires $de > 0$, so the conventional result reverses. The sign of $de/d\tau$ is therefore the opposite of the sign of $\varepsilon_X + \varepsilon_M - 1$. This is why the condition is both necessary and sufficient.

The Cobb–Douglas case. With constant expenditure shares, import volume is $M = \alpha_{T_B} I/p$, so $\varepsilon_M = 1$; similarly $\varepsilon_X = 1$. Then $\varepsilon_X + \varepsilon_M = 2 > 1$ and (21) is positive. Marshall–Lerner holds automatically, which is why the closed form (10) is monotone in τ.

Multilateral remark. In the three-country model of Section 2, each bilateral relationship satisfies a Marshall–Lerner condition, and the tariff still improves A's overall balance at fixed exchange rates. The bilateral conditions do not determine how the required adjustment is split between e_{AB} and e_{AC}. That split depends on substitution across foreign origins, governed by ρ, relative to substitution between home and foreign goods, governed by η. This is the subject of Section 2.

2 Three-country nested model

2.1 Setup

Countries $i \in \{A, B, C\}$, each producing tradable T_i and non-tradable N_i with the same linear technology, wage equation, and normalization as (2). Exchange rates e_{ij} (i's currency per unit of j's), $e_{ii} = 1$, no-arbitrage $e_{ij} = e_{ik}e_{kj}$; the free rates are e_{AB}, e_{AC}. Tariffs $\tau_{ij} \geq 0$ (i's tariff on j's good). Consumer price of T_j in i: $p_{ij} = (1 + \tau_{ij})\, e_{ij}$.

Preferences (two-layer nest):

$$C_i = C_{T_i}^{\alpha_T}\, C_{N_i}^{1-\alpha_T} \qquad \text{(Cobb–Douglas outer)} \tag{24}$$

$$C_{T_i} = \left[\alpha_D^{1/\eta}\, C_{T_i i}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta}\, M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \qquad \text{(home vs. imports, elasticity } \eta) \tag{25}$$

$$M_i = \left[\textstyle\sum_{j \neq i} \beta_j^{1/\rho}\, C_{T_j i}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \qquad \text{(across origins, elasticity } \rho) \tag{26}$$

with home weight α_D, origin weights β_j (symmetric case: $\beta_j = 1/2$). The single-elasticity model is the special case $\rho = \eta$.

2.2 Price indices and expenditure shares

CES price indices (in i's currency):

$$P_{M_i} = \left[\textstyle\sum_{j \neq i} \beta_j\, p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \qquad P_{T_i} = \left[\alpha_D\, p_{ii}^{1-\eta} + (1-\alpha_D)\, P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}. \tag{27}$$

Expenditure shares of total income I_i (rows sum to α_T):

$$s_{ii} = \alpha_T\, \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \tag{28}$$

$$s_{ij} = \alpha_T\, (1-\alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_j \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \qquad j \neq i. \tag{29}$$

2.3 Income and trade balances

Tariff revenue is rebated lump-sum. A fraction $\tau_{ij}/(1+\tau_{ij})$ of spending $s_{ij}I_i$ is revenue, so

$$I_i = w_i L_i + \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij} I_i \quad \Longrightarrow \quad I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}}\, s_{ij}}. \tag{30}$$

The shares depend only on prices, so (30) is a closed form.

Trade balances at producer prices (spending $s_{ij}I_i$ is tariff-inclusive; divide by $1+\tau_{ij}$ to get the producer-price value):

$$\mathrm{TB}_i = \sum_{k\neq i} \frac{s_{ki}\, I_k}{1+\tau_{ki}} - \sum_{j\neq i} \frac{s_{ij}\, I_i}{1+\tau_{ij}}, \tag{31}$$

with all quantities expressed in a common currency (this does not affect the zeros). Equilibrium: $\mathrm{TB}_B = \mathrm{TB}_C = 0$; $\mathrm{TB}_A = 0$ follows by Walras' law.

2.4 Linearization at the symmetric point

Set $L_i = 1$, $A_{T_i} = A_{N_i} = 1$, $\beta_j = 1/2$, and evaluate at free trade ($\tau_{ij} = 0$), where $e_{AB} = e_{AC} = 1$ and all incomes equal 1. Baseline shares:

$$s_{ii} = \alpha_T \alpha_D \equiv a, \qquad s_{ij} = \tfrac{1}{2}\,\alpha_T(1-\alpha_D) \equiv \tfrac{m}{2}, \qquad m \equiv \alpha_T(1-\alpha_D). \tag{32}$$

Perturb by $d\tau$ (A's tariff on B) and let $\nu_j \equiv d\log e_{Aj}$ ($\nu_A = 0$). Log price changes: $d\log p_{ij} = \nu_j + \mathbf{1}\{(i,j) = (A,B)\}\, d\tau$.

Index changes, from (27):

$$d\log P_{M_i} = \sum_{j\neq i} \beta_j\, d\log p_{ij}, \qquad d\log P_{T_i} = \alpha_D\, d\log p_{ii} + (1-\alpha_D)\, d\log P_{M_i}. \tag{33}$$

Explicitly:

$$d\log P_{M_A} = \tfrac{1}{2}(\nu_B + \nu_C + d\tau), \qquad d\log P_{M_B} = \tfrac{1}{2}\nu_C, \qquad d\log P_{M_C} = \tfrac{1}{2}\nu_B. \tag{34}$$

Share changes, from (28)–(29):

$$d\log s_{ii} = (1-\eta)\big(d\log p_{ii} - d\log P_{T_i}\big), \tag{35}$$
$$d\log s_{ij} = (1-\rho)\big(d\log p_{ij} - d\log P_{M_i}\big) + (1-\eta)\big(d\log P_{M_i} - d\log P_{T_i}\big). \tag{36}$$

Income changes, from (30) (the revenue term is first order for country A only):

$$d\log I_A = \tfrac{m}{2}\, d\tau, \qquad d\log I_B = \nu_B, \qquad d\log I_C = \nu_C. \tag{37}$$

Differentiate (31) for $i = B, C$ at the baseline. Each term $s\, I/(1+\tau)$ contributes $\frac{m}{2}\,(d\log s +$

$d\log I - d\tau_\cdot$), where $d\tau_\cdot = d\tau$ appears only in the A-to-B flow:

$$d\,\mathrm{TB}_B = \tfrac{m}{2}\Big[\big(d\log s_{AB} + \tfrac{m}{2}d\tau - d\tau\big) + \big(d\log s_{CB} + \nu_C\big) - \big(d\log s_{BA} + \nu_B\big) - \big(d\log s_{BC} + \nu_B\big)\Big] = 0, \tag{38}$$

$$d\,\mathrm{TB}_C = \tfrac{m}{2}\Big[\big(d\log s_{AC} + \tfrac{m}{2}d\tau\big) + \big(d\log s_{BC} + \nu_B\big) - \big(d\log s_{CA} + \nu_C\big) - \big(d\log s_{CB} + \nu_C\big)\Big] = 0. \tag{39}$$

Substituting (35)–(36) and collecting terms in $(\nu_B, \nu_C, d\tau)$ gives a linear system. Define

$$D \equiv 3\alpha_D(\eta - 1) + \rho + 1, \qquad \rho^* \equiv 3\big[\,1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)\,\big]. \tag{40}$$

Collecting terms yields the linear system

$$-\frac{mD}{4}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \frac{m}{4}\begin{pmatrix} \rho + \rho^*/3 \\ -\rho + \rho^*/3 \end{pmatrix} d\tau. \tag{41}$$

Invert using $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$:

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = -\frac{1}{3D}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} \rho + \rho^*/3 \\ -\rho + \rho^*/3 \end{pmatrix} d\tau = \frac{1}{3D}\begin{pmatrix} -(\rho + \rho^*) \\ \rho - \rho^* \end{pmatrix} d\tau. \tag{42}$$

Result 2.1 (Threshold)**.** *At the symmetric point,*

$$\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho + \rho^*}{3D} < 0, \qquad \boxed{\left.\frac{d\log e_{AC}}{d\tau}\right|_{\tau=0} = \frac{\rho - \rho^*}{3D}} \tag{43}$$

with $D > 0$ for $\eta \geq 1$ (local stability). A appreciates against B for all parameters; A depreciates against the untariffed C if and only if $\rho > \rho^*$.

2.5 Impossibility under a single elasticity

Impose $\rho = \eta = \sigma$ (the single-elasticity CES). Then

$$\rho^* - \sigma = 3 + 3\alpha_D(\sigma - 1) - 3\alpha_T(1 - \alpha_D) - \sigma = \sigma(3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T). \tag{44}$$

Result 2.2 (Impossibility)**.** *If $\alpha_D \geq 1/3$, both terms in* (44) *are nonnegative and the second is positive, so $\rho^* > \sigma$ for every $\sigma > 0$: the conventional appreciation against C survives at any elasticity. At $\alpha_D = 1/3$, $\alpha_T = 3/4$ (the symmetric baseline calibration $\alpha_{T_j} = \alpha_N = 1/4$): $\rho^* - \sigma = 1/2$ and $D = 2\sigma$, so*

$$\left.\frac{d\log e_{AC}}{d\tau}\right|_{\tau=0} = \frac{\sigma - \rho^*}{3D} = -\frac{1}{12\sigma} \;\longrightarrow\; 0^-. \tag{45}$$

2.6 Nominal versus real exchange rate in the three-country model

The quantities of Section 1.5 carry over: e_{Aj} is proportional to j's relative wage in A's currency and to the inverse of A's bilateral terms of trade with j. The real rate against C is $q_{AC} = e_{AC} P_C / P_A$ with $P_i = \kappa_i \, (P_{T_i}^{\text{own}})^{\alpha_T} (P_{N_i})^{1-\alpha_T}$, where $P_{T_i}^{\text{own}}$ is the tariff-inclusive tradable index (27) in i's own currency and $P_{N_i} = A_{T_i}/A_{N_i}$ is constant. Hence

$$d \log q_{AC} = d \log e_{AC} + \alpha_T \big(d \log P_{T_C}^{\text{own}} - d \log P_{T_A}^{\text{own}} \big). \tag{46}$$

At the symmetric point, converting the indices to own currency ($d \log P_{T_C}^{\text{own}} = d \log P_{T_C} - \nu_C$):

$$d \log P_{T_A}^{\text{own}} = (1 - \alpha_D) \, \tfrac{1}{2} (\nu_B + \nu_C + d\tau), \tag{47}$$

$$d \log P_{T_C}^{\text{own}} = \alpha_D \nu_C + (1 - \alpha_D) \tfrac{1}{2} \nu_B - \nu_C = (1 - \alpha_D) \big(\tfrac{1}{2} \nu_B - \nu_C \big). \tag{48}$$

Substituting and using $m = \alpha_T (1 - \alpha_D)$:

$$d \log q_{AC} = \nu_C + m \Big(\tfrac{1}{2} \nu_B - \nu_C - \tfrac{1}{2} \nu_B - \tfrac{1}{2} \nu_C - \tfrac{1}{2} d\tau \Big) = \Big(1 - \tfrac{3m}{2} \Big) \nu_C \; - \; \tfrac{m}{2} \, d\tau. \tag{49}$$

As in (17), the tariff appreciates A in real terms directly through the price index (the $-\frac{m}{2} d\tau$ term), and the pass-through from the nominal rate is attenuated. Setting (49) to zero with $\nu_C = \frac{\rho - \rho^*}{3D} \, d\tau$ gives the real-rate reversal threshold: $(2 - 3m)(\rho - \rho^*) = 3mD$, and since $D = 3\alpha_D (\eta - 1) + \rho + 1$ this is linear in ρ, with solution

$$\boxed{\rho_q^* = \frac{(2 - 3m) \, \rho^* + 3m \big[3\alpha_D (\eta - 1) + 1 \big]}{2 - 6m}} \qquad (\rho_q^* > \rho^* \text{ for } m \in (0, \tfrac{1}{3})). \tag{50}$$

At the benchmark ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, so $m = 0.08$): $\rho^* = 3.96$ and $\rho_q^* = 4.93$. Real depreciation against C requires more diversion than nominal depreciation, because part of the nominal movement is absorbed by the tariff's direct effect on A's price index.

3 The N-country case

3.1 General linear structure

Countries $i \in \{1, \ldots, N\}$, country 1 the numeraire, with the same technology and nested preferences as Section 2: home share α_{D_i}, origin weights β_{ij}, elasticities (η, ρ). Let $\nu = (\nu_2, \ldots, \nu_N)$ collect the log exchange rates and let a tariff $d\tau$ be imposed by country 1 on country 2. Linearizing the $N - 1$ independent trade-balance conditions at a free-trade baseline gives

$$J \, d\nu = -F \, d\tau \quad \Longrightarrow \quad \frac{d\nu}{d\tau} = (-J)^{-1} F = \frac{\operatorname{adj}(-J) \, F}{\det(-J)}, \tag{51}$$

where $J_{ij} = \partial \mathrm{TB}_i / \partial \nu_j$ and $F_i = \partial \mathrm{TB}_i / \partial \tau$ at fixed exchange rates (the impact effects). The entries of J and F are linear in η and ρ, with coefficients given by baseline expenditure shares and incomes, exactly as in the three-country linearization of Section 2.4.

Stability. The adjustment process $\dot{\nu}_i \propto \mathrm{TB}_i$ is locally stable if and only if all eigenvalues of J have negative real parts. Any such matrix satisfies $\det(-J) > 0$, so at a stable equilibrium the common denominator in (51) is positive and the sign of each response equals the sign of its numerator.

The multilateral Marshall–Lerner condition. Suppose the demand system satisfies gross substitutability. Then J has negative diagonal entries (each country's own appreciation worsens its own balance: the bilateral Marshall–Lerner conditions), nonnegative off-diagonal entries (another country's appreciation improves your balance), and a dominant diagonal (from homogeneity and Walras' law, with the numeraire absorbing the residual). A matrix with this sign pattern is a stable Metzler matrix, and $-J$ is then an M-matrix. M-matrices are inverse-positive:

$$(-J)^{-1} \;\geq\; 0 \quad \text{elementwise.} \tag{52}$$

Result 3.1 (Generic sign rule)**.** *At a stable equilibrium satisfying the multilateral Marshall–Lerner condition,*

$$\operatorname{sign}\!\left(\frac{d\nu_i}{d\tau}\right) = \operatorname{sign}\!\Big(\sum_j w_{ij} F_j\Big), \qquad w_{ij} \equiv \big[(-J)^{-1}\big]_{ij} > 0. \tag{53}$$

In particular, if all impact effects share a sign, all responses share it. Sign ambiguity arises only when the impact vector is mixed.

Because F is linear in ρ and $\operatorname{adj}(-J)$ is a polynomial of degree $N-2$ in ρ, each numerator in (51) is a polynomial of degree $N-1$ in ρ. At $N = 3$ this is a quadratic with one relevant root, which yields the scalar threshold of Section 2. At $N \geq 4$ single crossing is not guaranteed: there exist asymmetric configurations in which the sign of a bystander's response changes twice as ρ rises, so no scalar threshold exists in general. The sign of any particular response remains computable from (53).

3.2 The symmetric case: a closed form

Let all countries have equal size, common preferences, home share α_D, and equal origin weights $1/(N-1)$. Country 1 (A) tariffs country 2 (B); the remaining $k = N-2$ bystanders are identical, with representative c. By symmetry the system (51) reduces to two unknowns, (ν_B, ν_C). Define

$$\rho_1^* \;\equiv\; 1 + \alpha_D(\eta - 1) - \alpha_T(1-\alpha_D), \qquad D_N \;\equiv\; N\alpha_D(\eta-1) + (N-2)\rho + 1. \tag{54}$$

Solving the two-equation system (the derivation parallels Section 2.4 and is verified symbolically in the replication package) gives

$$\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{N\rho_1^* + (N-2)\rho}{N\,D_N} \;<\; 0, \tag{55}$$

$$\left.\frac{d\log e_{AC}}{d\tau}\right|_{\tau=0} = \frac{\rho - N\rho_1^*}{N\,D_N}, \tag{56}$$

$$\left.\frac{d\log e_{BC}}{d\tau}\right|_{\tau=0} = \frac{(N-1)\,\rho}{N\,D_N} \;>\; 0, \tag{57}$$

with $D_N > 0$ the stability condition. Hence:

Result 3.2 (N-country threshold)**.** *In the symmetric N-country model, an isolated tariff depreciates A's currency against each untariffed bystander if and only if*

$$\boxed{\rho \;>\; \rho^*(N) \;=\; N\big[\,1 + \alpha_D(\eta-1) - \alpha_T(1-\alpha_D)\,\big]} \tag{58}$$

At $N = 3$ this is the threshold of Section 2; the coefficient 3 there is the number of countries. The threshold grows linearly in N: with symmetric weights, each bystander receives a share $1/(N-1)$ of the diverted expenditure, so the substitutability required for any single bystander to gain scales with the number of countries.

For the symmetric trade war, add the responses to $d\tau_{AB}$ and its mirror $d\tau_{BA}$. Relabeling countries gives $\nu_C^{\text{war}} = 2\nu_C - \nu_B$, and substituting (55)–(56):

$$\left.\frac{d\log e_{AC}}{d\tau}\right|_{\tau=0}^{\text{war}} = \frac{\rho - \rho_1^*}{D_N}, \qquad \left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0}^{\text{war}} = 0. \tag{59}$$

Result 3.3 (War threshold is independent of N)**.** *In a symmetric trade war, every bystander's currency appreciates against both belligerents if and only if*

$$\boxed{\rho \;>\; \rho_1^* \;=\; \frac{\rho^*(N)}{N}} \tag{60}$$

for any number of countries. The reason is that the belligerents' adjustments offset (e_{AB} is unchanged), so each bystander's equation decouples and its sign equals the sign of its own impact effect, which is a per-country condition that does not dilute with N. At $\eta = 1$, $\rho_1^* = 1 - \alpha_T(1-\alpha_D) < 1$, so any $\rho \geq 1$ produces joint depreciation of the belligerents; for $\eta > 1$, ρ_1^* can exceed one, and sufficiently low cross-origin substitutability reverses even the war case.

The formulas above are verified against nonlinear equilibrium solves at $N = 4$ (threshold $\rho^*(4) = 5.28$ at $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, matching the numerical root to five decimal places).

4 Canonical PTA model: competing exporters

4.1 Setup

Countries A (importer), B and C (exporters). Two goods: x and numeraire y. Endowments: A holds only y; B and C hold x (and possibly y). Quasilinear preferences in each country,

$$U_i = u_i(x_i) + y_i, \qquad u_i' > 0,\ u_i'' < 0. \tag{61}$$

A levies specific tariffs t_B, t_C on imports of x from B and C. Let q_j be the price received by exporter j (the world price of its good) and p the domestic price of x in A. Both origins sell in A at the same domestic price:

$$p = q_B + t_B = q_C + t_C. \tag{62}$$

4.2 Demands and market clearing

From $u_i'(x_i) =$ (local price of x):

$$M_A(p) = x_A^d(p), \qquad E_j(q_j) = \omega_j - x_j^d(q_j), \quad j = B, C, \tag{63}$$

with $M_A' < 0$ and $E_j' > 0$. Market clearing for x:

$$M_A(p) = E_B(q_B) + E_C(q_C) = E_B(p - t_B) + E_C(p - t_C). \tag{64}$$

Equation (64) determines p given (t_B, t_C); then q_B, q_C follow from (62).

4.3 Comparative statics of a discriminatory tariff

Totally differentiate (64) with respect to t_B, holding t_C:

$$M_A' \frac{dp}{dt_B} = E_B'\Big(\frac{dp}{dt_B} - 1\Big) + E_C' \frac{dp}{dt_B} \quad \Longrightarrow \quad \frac{dp}{dt_B} = \frac{E_B'}{E_B' + E_C' - M_A'} \in (0,1). \tag{65}$$

Therefore

$$\frac{dq_B}{dt_B} = \frac{dp}{dt_B} - 1 = \frac{E_C' - M_A'}{E_B' + E_C' - M_A'} \cdot (-1) \ < \ 0, \tag{66}$$

$$\frac{dq_C}{dt_B} = \frac{dp}{dt_B} \ > \ 0. \tag{67}$$

Result 4.1 (Viner–Mundell). *A discriminatory tariff on B worsens B's terms of trade and improves C's terms of trade. Equivalently, a preferential tariff cut toward B (lower t_B) improves B's and worsens C's terms of trade.*

4.4 Relation to the exchange-rate model

In the model of Section 2, an improvement in C's terms of trade corresponds to a rise in C's wage in common-currency units, that is, a rise in e_{AC}: A depreciates against C. The canonical PTA model delivers this sign unconditionally, from the slope conditions $E_j' > 0$ and $M_A' < 0$ alone, while the model of Section 2 requires $\rho > \rho^*$. The difference is accounted for by four ingredients, each of which removes one of the forces that oppose diversion.

(i) No home tradables: $\alpha_D = 0$. In (64), A produces no x, so tariffed demand can only switch between B and C; the home–import margin does not exist. In the notation of Section 2, $\alpha_D = 0$ and the term $\alpha_D(\eta - 1)$ drops from the threshold, leaving $\rho^* = 3(1 - \alpha_T)$.

(ii) No non-tradables: $\alpha_T = 1$. The residual good y in the PTA model is itself freely traded, so all expenditure is on traded goods. Setting $\alpha_T = 1$ in the remaining threshold gives $\rho^* = 0$: at the symmetric point of Section 2, full tradability and zero home bias already make any $\rho > 0$ sufficient. The remaining bar at $\alpha_T < 1$ reflects the non-tradable sector shrinking the divertible expenditure relative to the income drag operating through B.

(iii) No target–bystander linkages. In the competing-exporters structure, B and C sell x to A and buy y from A, but do not trade with each other, and quasilinear preferences remove income effects. Both forces that oppose diversion in Section 2 run through the B–C relationship: C substituting its spending toward B's cheapened goods, and B's income loss reducing its purchases from C. Neither can operate here. C's terms of trade are determined entirely in A's import market, where the tariff on B can only help C. This is why the PTA sign holds at any baseline, not only at a symmetric point: the opposing forces are absent structurally, not balanced away.

(iv) The Viner limit: $\rho = \infty$. In the original customs-union analysis, B and C export the same physical commodity, which is the limit of perfect substitutability across origins.

The canonical PTA result is therefore the limit of the Section 2 model as $\alpha_D \to 0$, $\alpha_T \to 1$, the target–bystander linkages are removed, and (in the strictest version) $\rho \to \infty$. Each feature restored — home tradables, non-tradables, triangular trade with income effects — adds one opposing claim on the sign of C's terms of trade, and the threshold ρ^* prices those claims.

4.5 Scope of the PTA literature for the exchange-rate question

The asymmetry of the canonical model — A acts, B and C are passive supply curves — is not special to this exposition. The analytical PTA literature obtains its sign results through some combination of the four ingredients above:

- *Competing-exporter and quasilinear models* (Bagwell–Staiger 1999 and the GATT literature; Ornelas 2005 with linear oligopoly): ingredient (iii). Exporters do not trade with each other, and quasilinearity or partial equilibrium removes income effects.

- *Symmetric endowment-bloc models* (Krugman 1991; Bond–Syropoulos 1996): ingredients (i) and (ii). Every good is traded, each country is fully specialized with no distinct home-consumption margin, and a single elasticity governs all substitution — the flat-CES, all-tradable world in which Section 2's thresholds are low or absent by construction.

- *The Viner tradition*: ingredient (iv), identical export goods.

- *Kemp–Wan 1976* is general in preferences and country number but proves existence of a compensating external tariff; it makes no comparative-statics sign claims. Where the literature becomes general, it stops signing.

Within the analytical literature the pattern is consistent: unambiguous third-country signs are purchased by removing the margins that make the exchange-rate question interesting, and where the structure is generalized, the sign claims are conditioned (Mundell's results assume gross substitutability; Lipsey's amendment to Viner shows that richer demand structure alone makes welfare conclusions ambiguous) or withdrawn (Kemp–Wan).

Models with the offsetting margins did exist, on the applied side. Computable general equilibrium models of preferential agreements in the GTAP tradition use, as standard practice, a two-level Armington nest — one elasticity between domestic goods and the import composite and a larger one across import sources — together with full trade matrices, non-tradables, and income effects. Modern quantitative trade models are in the same position, with the two-elasticity version estimated by Li et al. Excluded-country terms-of-trade effects in these models can take either sign, configuration by configuration. But this end of the literature reports simulation output; it has not asked *when* the sign flips.

The division of labor is therefore: the analytical PTA literature establishes the possibility of the result — a discriminatory tariff improving the excluded country's terms of trade, hence depreciating the tariffer against it — with the opposing forces removed by construction; the applied literature contains the opposing forces but has not characterized the sign; and the condition itself was stated in neither terms-of-trade nor exchange-rate language. The nested three-country model is the minimal setting that both nests the analytical result (as the limit of the previous subsection) and locates the threshold that the applied structures contain implicitly.